

Tuan Nguyen (Mike)

Machine Learning & Data Engineer

Sydney, NSW · tuannm3812@gmail.com · tuannm3812.github.io · github.com/tuannm3812 · linkedin.com/in/tuan-m-nguyen · kaggle.com/tuannm3812

PROFESSIONAL SUMMARY

Machine Learning and Data Engineer with **7+ years of experience** building predictive models, data pipelines, and deployable AI applications. Proven track record designing stateful conversational agents, schema-based retrieval systems, and full-stack ML prototypes. Expert in Python backend services and intermediate in the **TypeScript stack (React, Node.js/Express)**. Developed flagship wellness/nutrition AI products utilizing multi-agent orchestration, local RAG pipelines, API integrations, and robust validation guardrails.

TECHNICAL SKILLS

Languages & Core: Python, SQL, TypeScript, Shell Scripting (Bash), HTML/CSS, Git, unit testing (pytest)

Data & Cloud: Google BigQuery, Snowflake, Azure Databricks, Delta Lake, Unity Catalog, Apache Airflow, dbt, PostgreSQL, ETL/ELT pipelines, Medallion Architecture (Bronze-Silver-Gold)

ML & Deep Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, computer vision, time-series forecasting, hyperparameter tuning (Optuna, Hyperopt)

AI & Agent Engineering: LangGraph, LangChain, multi-agent systems, schema-driven Pydantic outputs, prompt engineering, RAG pipelines, vector databases, local LLMs (Ollama)

Backend & Productization: FastAPI, Streamlit, Node.js, Express, React, CI/CD (GitHub Actions), Docker, Looker Studio

PROFESSIONAL EXPERIENCE

Shopee Vietnam

Dec 2021 – Jan 2025

Lead Data Optimization & Insights Manager | Growth & Acquisition

- **Engineered** Python predictive models (churn forecasting, K-means clustering) in BigQuery to optimize distribution of growth vouchers, **driving an 85%+ partner milestone graduation rate**.
- **Architected** BigQuery ETL pipelines consolidating transaction and seller traffic datasets, **reducing manual data-processing latency by 80%** and decreasing partner onboarding drop-off by 25%.
- **Developed** regression and econometric models measuring partner fee elasticity at shop level, **improving voucher spend efficiency and investment ROI by 15%**.
- **Automated** lead scraping pipelines using Python (BeautifulSoup, REST APIs) to consolidate competitor seller data, replacing offline tracking spreadsheets.
- **Coached** an analytics team of 5 data professionals on SQL, predictive modeling, and automated report delivery in Looker Studio.

FPT Software & FPT Digital (FDX)

Dec 2020 – Dec 2021

Strategy Lead (Global Affairs) & Digital Transformation Consultant

- **Audited** digital maturity and designed cloud-migration data blueprints for enterprise conglomerates (e.g. Minh Phu Seafood Corp, Vietnam Electricity) in collaboration with McKinsey and BCG.
- **Led** Salesforce CRM implementation workstreams and delivered predictive analytics modules for lead scoring, **driving a 20% improvement in lead-to-opportunity conversion**.
- **Negotiated** technology partnership frameworks with multinational corporations, **securing FPT as the local systems integrator** for foreign direct investment clients.
- **Translated** client operational bottlenecks into implementation roadmaps, agile delivery plans, and Customer Data Platform (CDP) architecture recommendations.

- **Directed** financial audit engagements as Senior-in-Charge for a portfolio of 40+ commercial banks, securities firms, investment funds, and corporate clients.
- **Programmed** Python (pandas) and SQL data-validation scripts on client Oracle databases, **automating bank reconciliation workflows and cutting execution time by 90%** (2 days to under 1 hour).
- **Analyzed** core banking transaction ledgers using SQL to identify security gaps and double-posted entries, **expanding audit coverage to 100% of transaction data**.
- **Presented** database control deficiencies and systemic booking anomalies to client C-level executives, driving database remediation strategies.

SELECTED PROJECTS

AI Meal Planner — AI Product / Full Stack

FastAPI · Streamlit · scikit-learn · Local RAG · Pydantic · USDA API · CI

- Architected a **multi-agent planning workflow** with FastAPI and Pydantic, executing parallelized dietary constraint checks and USDA/FatSecret nutrition lookups.
- Enforced **type-safe contract validation** under automated CI testing, reducing client-server schema mismatch errors on meal suggestion payloads.

github.com/tuannm3812/ai-meal-planner · [Live demo](#)

Enterprise Text-to-SQL Agent — Applied AI / RAG

Gemini/Ollama · Schema RAG · Hybrid Retrieval · SQLGlot · SQLite · Streamlit

- Built a **secure schema-RAG Text-to-SQL agent** with query-repair loops, achieving high-fidelity translations without exposing raw client database records to LLM API endpoints.
- Enforced **database execution safety** using SQLite authorizer controls and SQLGlot AST validation to block prompt injections and unauthorized DDL commands.

github.com/tuannm3812/aipa-text-to-sql-agent · [Live demo](#)

Production-Grade ELT Pipeline: Airbnb Market Analytics — Data Engineering

Python · Airflow · dbt · PostgreSQL · Medallion Architecture · SCD Type 2

- Designed a **Medallion-architecture ELT pipeline** for Sydney Airbnb and Census logs, automating historical dimensional snapshots via dbt SCD Type 2.
- Orchestrated **monthly data load schedules** using Apache Airflow, managing schema evolution and reducing data drift errors across LGA regional dimensions.

github.com/tuannm3812/airbnb-ELT-warehouse

NYC Taxi Databricks — Data Engineering / Lakehouse

PySpark · Databricks · Delta Lake · Spark SQL · Ridge Regression

- Built a **Databricks lakehouse workflow** for large-scale NYC taxi trip analytics, borough-level query segmentation, and fare prediction modeling.
- Managed **Delta Lake table curation** (Bronze-Silver-Gold), PySpark cleaning, schema harmonization, trip feature engineering, and regression diagnostics.

github.com/tuannm3812/NYC-Taxi-Databricks

EDUCATION

Master of Data Science and Innovation (High Distinction) — University of Technology Sydney *Expected Feb 2027*

Bachelor of International Economics and Business (Distinction) — Foreign Trade University *Jun 2017*